Near-Null Magnetic Fields in Lunar Plant Biology: The First 5 Days of Growth

1. LUNAR MAGNETIC REALITY

- Moon lacks global core dynamo
- Artemis South Pole: < 0.8 nT (Deep near-null magnetic void)
- Microclimatic field variation

Payload Field Architecture:

Btotal = Bcrust + Bext + Blander

Lander Avionics (50–500 μT)
Swamps Ambient Lunar HMF
⟶ Strict Shielding Required

Core Concept 1:
Physical Reality of Lunar HMF

2. FIRST 5 DAYS OF GROWTH

- 0–72 h Heterotrophic Bottleneck
- Peroxisomal beta-oxidation
- Mitochondrial restart burst
- Plastid greening transition

Empirical Phenotypes in NNMF:

- Viability: -20% (# Seedlings)
- Velocity: Δt50 = +16 h Lag
- Dry Biomass: -28.5% Loss
- Pigments: HEMA1, CHLH, POR & HY5 Repressed (-1.8 to -2.5x)

Core Concepts 2, 3, 4:
Viability, Speed, Pigments

3. MULTI-STRESSOR NEXUS

- Exploration Atmosphere (8.2 psia)
- Elevated CO2 (1200–3000 ppm)
- GCR Ionizing Radiation (1 Gy)
- Shared organellar bioenergetics

Mitochondrial Hub & Flight Action:

- Coregulated: AOX1D & CAT3 (+)
- Cross-Kingdom: Parallels astronaut mitochondrial stress
- Flight Operations: Predeclared Gates (Green / Amber / Red)

Core Concept 5:
CO2 ↔ O2 ↔ ROS Nexus